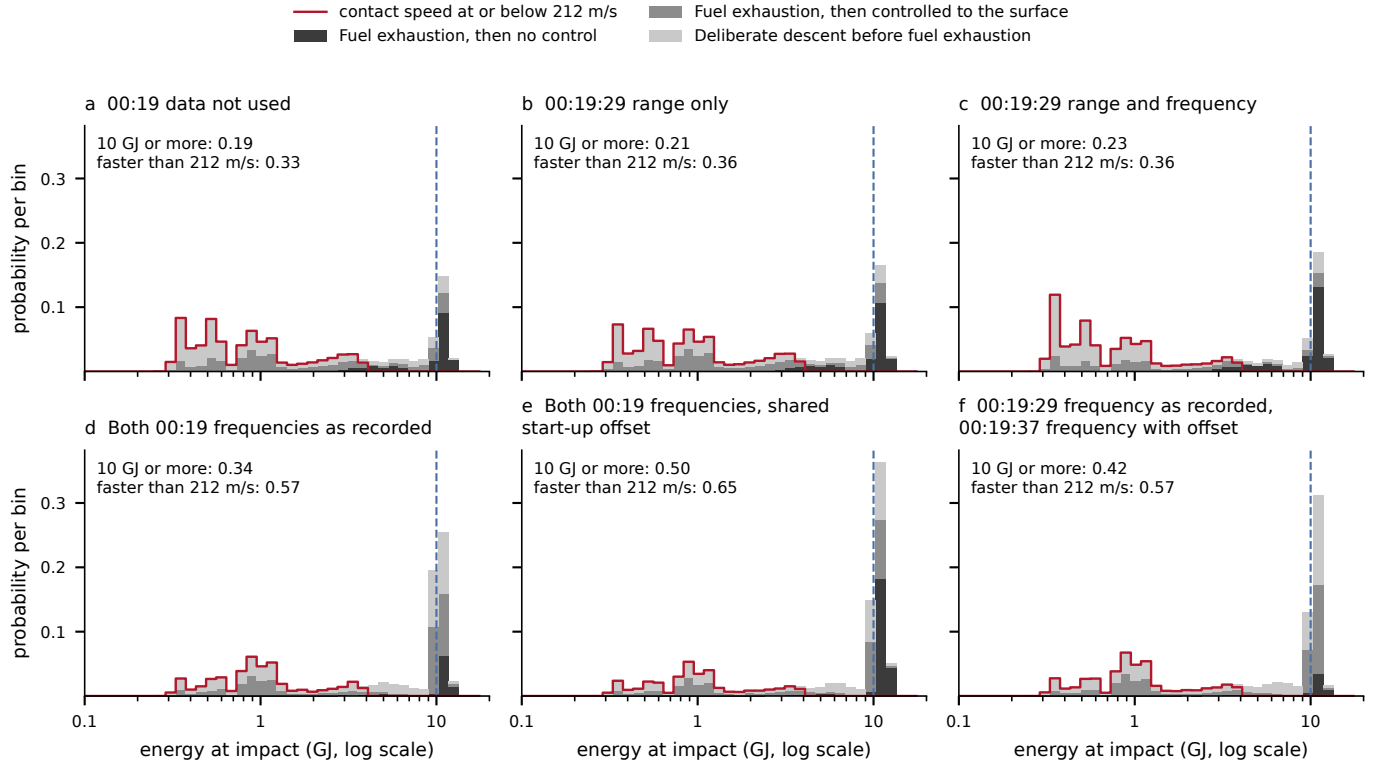
Half of the impacts carry 10 GJ or more when both 00:19 frequencies are used with a shared start-up offset; every impact of 10 GJ or more is faster than 212 m/s



Note 1: Each panel shows how the energy at impact is spread, for one use of the 00:19 satellite data. The grey shades show the type of flight end. The red line shows the part of the probability with a contact speed at or below 212 m/s (1.25 times the maximum operating speed). The impacts near 10-12 GJ are steep dives at about 350 m/s. The flight model for a descent without control is not calibrated, so these energies are not physical predictions.

Note 2: NOT A PHYSICAL PREDICTION: The uncontrolled-arm dynamics are not calibrated: the Boeing leave-one-out gate failed in all three rounds. Impact energy, attitude, flight-path angle and energy-transfer duration from this arm are a labelled sensitivity on uncalibrated dynamics, not a physical prediction. End of flight on core run J (accepted as a pass by project ruling, 16 Oct 2026): 4 route families with 24/18/16/16 seeds x 100,000 hand-off particles at 00:11; 16 children x 4 descents per particle (waypoints), 8 x 4 (others); family weights 0.9982/0.0013/0.0003/0.0002; binary mh370-eof-rj1 (sha256 19031f41). Round-3 uncontrolled-descent laws; spiral bank limit uniform 52-90 deg; dive to surface p = 0.5. Log-on causes C1 fuel exhaustion (Erlang lag) + C2 other (1/W, W = 00:10:58-01:15:56 UTC, silent), q = 1/3; survival factor (no in-flight breakup before 00:19:37) applied; transmission at 00:19 imposed. No treatment uses the 00:19:37 range. Treatments: D0' held out, D1 R600 BTO, D2 R600 BTO+BFO, D3 both BFOs raw, D4 both BFOs + shared start-up offset (Holland), D5 R600 BFO raw + R1200 BFO with offset U[17,130] Hz. Families: A1 = code 1, A2 = code 2, B = codes 3+6 (B uncontrolled about 0.001). PROVISIONAL: piloted-turn model, INTERIM pitch-down. Energy = contact kinetic energy 0.5 m v^2 (tau90_v1 energy_transferred_j, the declared upper bound). 40 log bins 0.1-20 GJ; values clipped into the range. 212 m/s = 1.25 VMO (330 kt), the G13 credibility threshold (latent:contact_speed_feasible). Weighted mass-preserving resample, 10,000 rows per seed per treatment, all four strata.